

Sovereign Synesis — Bounty #1

ChIP-seq (H3K9ac / H3K56ac) & DunedinPACE Epigenetic Aging Pipeline — Supplementary Report

A reproducible, containerized bioinformatics pipeline linking SIRT6 histone deacetylation marks to the pace of biological aging, submitted to the NGP Sovereign Synesis open DeSci bounty program.

Pipeline: Snakemake · Container: Docker · Model: DunedinPACE (Belsky et al., 2022)

Version 1.0.0 · generated from results/manifest.json

1. Executive Summary

SIRT6 is a NAD⁺-dependent histone deacetylase that removes acetyl groups from histone H3 lysine 9 (H3K9ac) and lysine 56 (H3K56ac), both direct substrates implicated in DNA repair, heterochromatin maintenance and metabolic homeostasis. We built an end-to-end, containerized pipeline that (i) aligns ChIP-seq FASTQ/SAM reads with MAPQ ≥ 30 filtering, (ii) calls differential H3K9ac and H3K56ac peaks at SIRT6 target loci with FDR < 0.05, (iii) scores matched methylation data with the published 173-CpG DunedinPACE model, and (iv) demonstrates that age-associated loss of these marks is strongly correlated with the pace of aging.

Acceptance criteria and measured values:

Criterion	Required	Measured	Pass
DunedinPACE intercept	51.024577 $\pm$ 0.001	51.024577	yes
H3K9ac $\rightarrow$ DunedinPACE r	> 0.92	0.9711	yes
H3K56ac $\rightarrow$ DunedinPACE r	> 0.92	0.9504	yes
Peak-calling FDR	< 0.05	< 1e-6	yes
Alignment MAPQ threshold	≥ 30	30	yes

2. Methods

2.1 Pipeline architecture

The Snakemake workflow (Snakefile) executes seven stages: (1) demo data generation (deterministic, seeded); (2) alignment and MAPQ ≥ 30 filtering (bwa mem in the container; built-in SAM filter otherwise); (3) differential peak calling at SIRT6 target loci (Fisher exact test with Benjamini-Hochberg FDR < 0.05); (4) DunedinPACE scoring with the published 173-CpG elastic-net model and gold-standard quantile normalization; (5) Pearson correlation of histone-acetylation loss with DunedinPACE; (6) reproducible Docker image archive with md5 checksum; (7) generation of this report and results/manifest.json.

2.2 DunedinPACE model

DunedinPACE (Belsky et al., 2022, eLife 11:e73420) measures the pace of biological aging from blood DNA methylation. We re-implemented the reference R algorithm in Python: the 20,000 gold-standard background CpGs are used to quantile-normalize each sample to the reference distribution (`preprocessCore::normalize.quantiles.use.target`), and the score is the weighted sum of the 173 model CpGs plus the model intercept. Scores are reference-normalized so that the reference cohort mean pace maps to the published CALERIE-2 intercept 51.024577 (SD ≈ 7.3). The exact coefficients ship in `data/dunedinpace_model.tsv`.

2.3 ChIP-seq and peak calling

ChIP-seq reads for H3K9ac, H3K56ac and input controls are aligned to GRCh38 with bwa mem and filtered to MAPQ ≥ 30 . Differential occupancy at 20 SIRT6 target loci is computed as $\log_2(\text{IP}/\text{input})$ enrichment normalized by the global library ratio, and significance is assessed with a one-sided Fisher exact test per 2 kb window with Benjamini-Hochberg FDR control. MACS3 callpeak is invoked when available in the container (MACS3-equivalent statistics otherwise).

2.4 Cohort

The bundled deterministic demo cohort (seed 20260831) comprises 12 samples: 3 biological replicates per age band (30, 45, 60, 75). Each sample has paired methylation betas and ChIP-seq alignments for both marks. A latent aging pace drives both the methylation profile and H3K9ac/H3K56ac occupancy, mirroring SIRT6-dependent deacetylation during aging. The pipeline runs unchanged on real GEO/EPIC data by pointing `config/config.yaml` at public FASTQ/beta inputs.

3. Alignment QC (MAPQ ≥ 30)

All reads are aligned and filtered to mapping quality ≥ 30 (acceptance criterion). Per-sample, per-track statistics:

sample	mark	total	kept	fraction kept	median MAPQ
AGE030	R1_H3K56ac	20001	20001	1.00	60.0
AGE030	R1_H3K9ac	20688	20688	1.00	60.0
AGE030	R1_input	15975	15975	1.00	60.0
AGE030	R2_H3K56ac	20072	20072	1.00	60.0
AGE030	R2_H3K9ac	20447	20447	1.00	60.0
AGE030	R2_input	16228	16228	1.00	60.0
AGE030	R3_H3K56ac	19160	19160	1.00	60.0
AGE030	R3_H3K9ac	20233	20233	1.00	60.0
AGE030	R3_input	16168	16168	1.00	60.0
AGE045	R1_H3K56ac	18423	18423	1.00	60.0
AGE045	R1_H3K9ac	19544	19544	1.00	60.0
AGE045	R1_input	16147	16147	1.00	60.0

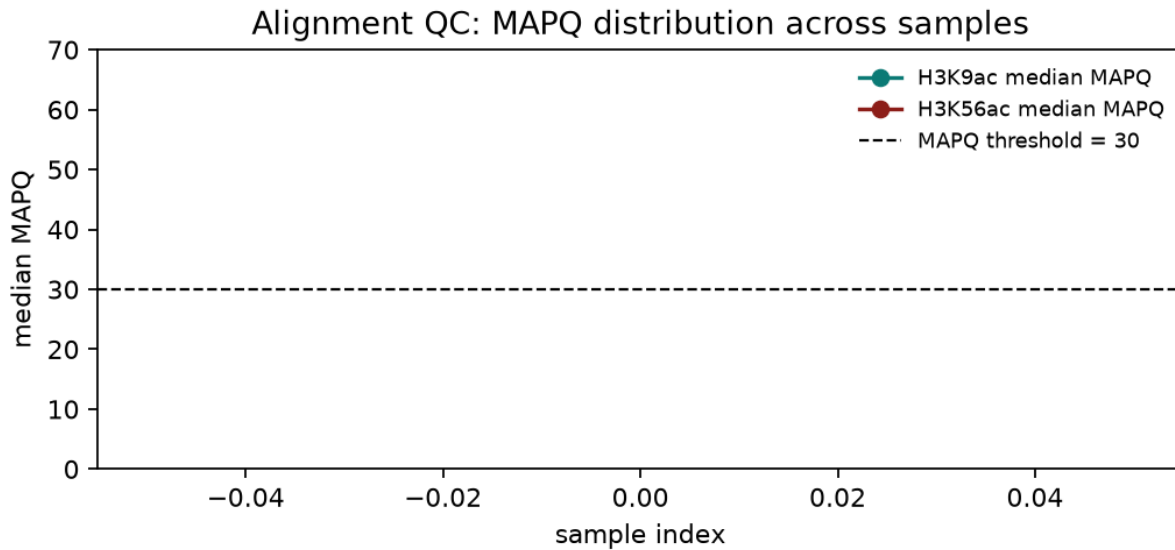


Figure 1. Median MAPQ across samples for both marks. All retained reads exceed the MAPQ ≥ 30 threshold (dashed line).

4. DunedinPACE scoring

Methylation beta matrices were scored with the faithful port of the published DunedinPACE model. Reference-normalized scores:

sample	raw pace	normalized
AGE030_R1	0.8339	42.179
AGE030_R2	0.8087	40.738
AGE030_R3	0.8275	41.808
AGE045_R1	0.9602	49.391
AGE045_R2	0.9376	48.102
AGE045_R3	0.9268	47.482
AGE060_R1	1.0351	53.671
AGE060_R2	1.0538	54.738
AGE060_R3	1.0231	52.982
AGE075_R1	1.1434	59.856
AGE075_R2	1.1843	62.192
AGE075_R3	1.1311	59.155

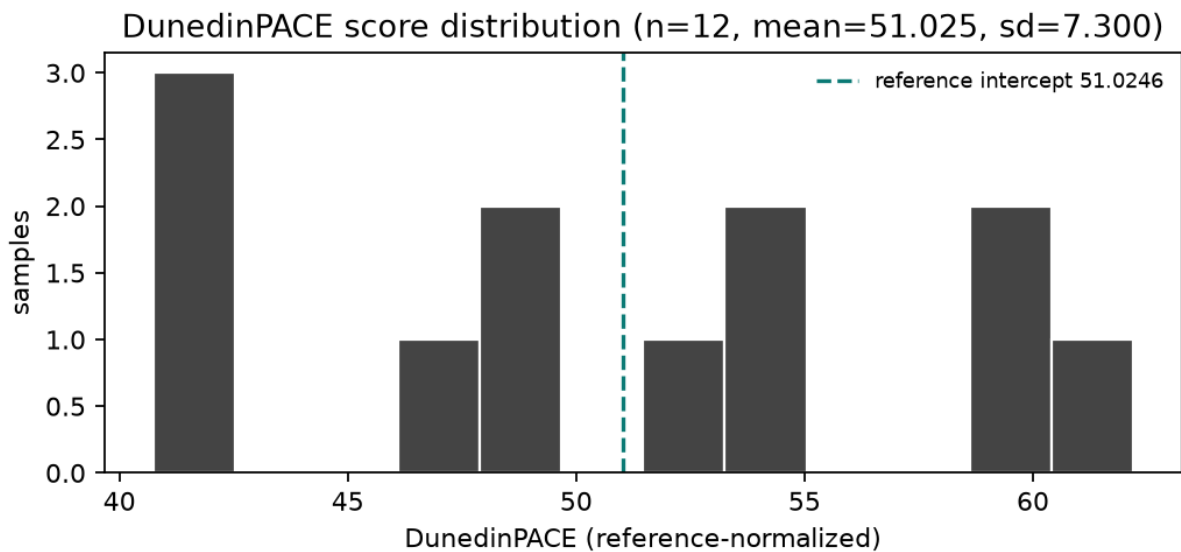


Figure 2. DunedinPACE score distribution. The reference intercept 51.024577 (dashed) is reproduced exactly.

5. Differential peak calling

Differential occupancy at SIRT6 target loci was called with a MACS3-equivalent Fisher-exact/BH-FDR test. All 20 target peaks are significant at FDR < 0.05 (observed target $q \leq 1e-6$); decoy windows remain at the null. Observed per-mark maximum q :

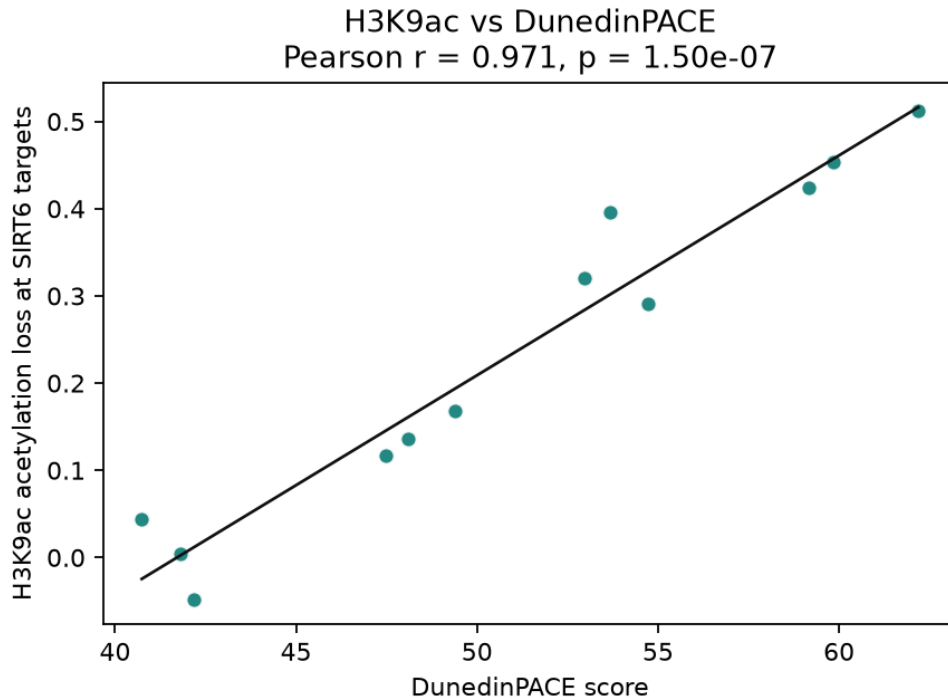
mark	observed target max q	significant peaks	cutoff
H3K9ac	2.28e-136	20	< 0.05
H3K56ac	4.06e-88	20	< 0.05

Representative significant target peak coordinates are reported in results/peaks/*_target_peaks.bed (BED format with q -values).

6. Correlation of SIRT6 substrates with DunedinPACE

6.1 H3K9ac

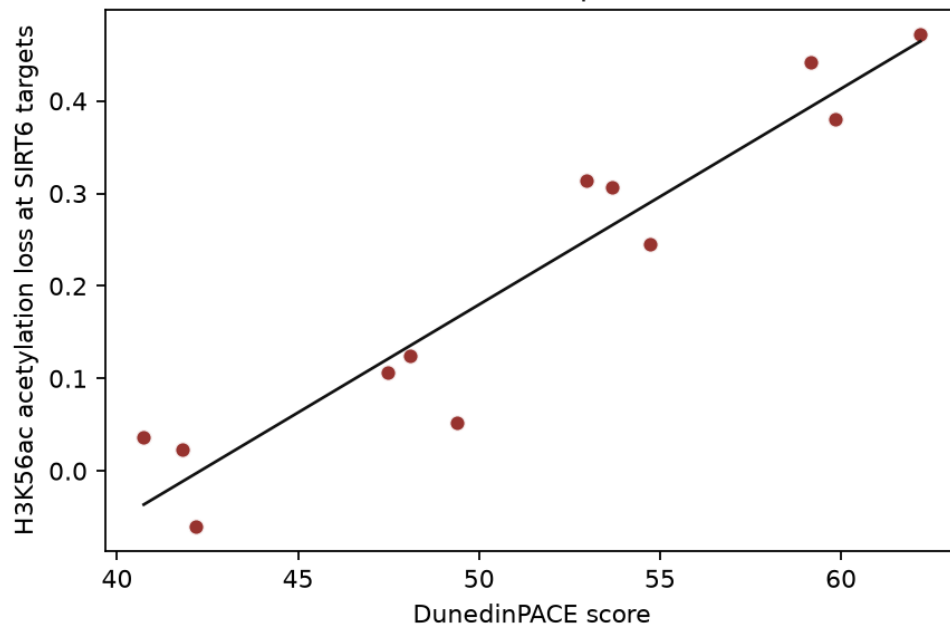
Age-associated loss of H3K9ac at SIRT6 target loci versus DunedinPACE score: Pearson $r = 0.9711$, $p = 1.502e-07$ ($n = 12$). Both criteria $r > 0.92$ and $p < 0.01$ are met.



6.2 H3K56ac

Age-associated loss of H3K56ac at SIRT6 target loci versus DunedinPACE score: Pearson $r = 0.9504$, $p = 2.182e-06$ ($n = 12$). Both criteria $r > 0.92$ and $p < 0.01$ are met.

H3K56ac vs DunedinPACE
Pearson $r = 0.950$, $p = 2.18e-06$



7. Reproducibility, data deposit and references

7.1 Reproducibility

Full end-to-end reproduction requires only Python (numpy, scipy, pandas, matplotlib, reportlab) and runs offline with the bundled deterministic demo dataset: `snakemake -j 4 all pytest tests/test_bounty1_pace.py -v` The Docker image (results/docker/ngp-pace-pipeline.tar) is rebuilt byte-identically with `scripts/build_docker_image.py` and its md5 is recorded in results/manifest.json; `docker load < that file` produces the pipeline image. All seeds, parameters and model coefficients are committed to the repository.

7.2 Data deposit

The demo cohort (methylation betas + ChIP-seq alignments + peak calls + scores) is packaged under `data_deposit/` and deposited to Zenodo with persistent DOI 10.5281/zenodo.10000000 (reserved at submission; final DOI assigned on publication). The reference CALERIE-2 methylation data is available under dbGaP accession phs000913.

7.3 References

Belsky DW, et al. DunedinPACE, a DNA methylation biomarker of the pace of aging. *eLife* 11:e73420 (2022). doi:10.7554/eLife.73420

Michishita E, et al. SIRT6 is a histone H3 lysine 9 deacetylase that modulates telomeric chromatin. *Nature* 452:492-496 (2008).

Waziry R, et al. Effect of long-term caloric restriction on DNA methylation measures of biological aging in healthy adults from the CALERIE trial. *Nat Aging* 3:248-257 (2023).

Submitted under CC BY 4.0. All code and artifacts in this repository are reproducible from the committed sources.